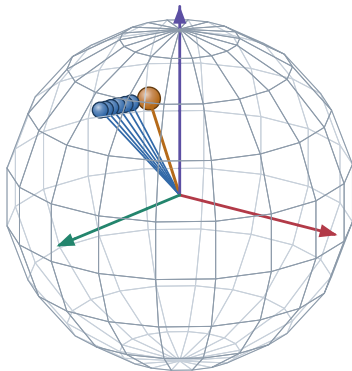


A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

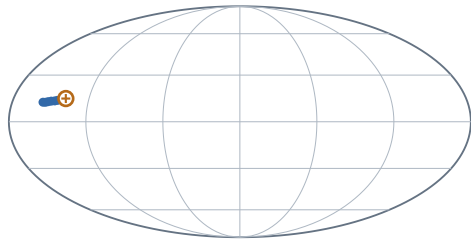


● logged history direction ⊕ example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



$+x_e$ at map centre · $+z_e$ north · wrap seam at $\pm 180^\circ$

$$\mathbf{M}_{\text{dir}(v)} = \sum_{k < t} w_{k(v)} \mathbf{d}_{k(v)} \mathbf{d}_{k(v)}^\top$$

$$\nu_{i(v)} = 1 - \frac{\mathbf{d}_i^\top \mathbf{M}_{\text{dir}} \mathbf{d}_i}{\text{tr } \mathbf{M}_{\text{dir}} + \varepsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\mathbf{M}_{\text{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.